

34.1 Exponential dispersion families

The normal linear model makes two assumptions usually stated together but logically independent: that $E(Y_i)$ is linear in the regressors, and that Y_i is normal with a variance not depending on that mean. This section replaces the second by the least it can, a one-parameter family whose log-density is linear in the data with a single extra scale. [Section 7.4](#) already wrote the normal linear model in that form (Eq. 7.4.1); what follows names the ingredients, because in a generalized linear model each does a separate job.

34.1.1 The family

Definition 34.1.1. Exponential dispersion family

Let ν be a σ -finite measure on $\mathbb{R}$. A random variable Y belongs to the **exponential dispersion family** generated by ν if, for a dispersion $\phi > 0$ and a known **prior weight** $w > 0$, its density with respect to ν is

$$f(y; \theta, \phi) = \exp\left\{\frac{w\{y\theta - b(\theta)\}}{\phi} + c(y, \phi, w)\right\}, \quad (34.1.1)$$

where θ lies in the **natural parameter space**

$$\Theta = \left\{\theta \in \mathbb{R} : \kappa(\theta) = \int e^{wy\theta/\phi} e^{c(y, \phi, w)} d\nu(y) < \infty\right\}$$

and $b(\theta) = (\phi/w) \log \kappa(\theta)$ is the **cumulant function**, required to be the *same* function of θ for every admissible pair (ϕ, w) . The normalizing constant $\kappa(\theta)$ is this chapter's only use of that letter with an argument; elsewhere $\kappa(\cdot)$ is the condition number of Definition 1.8.1. The parameter θ is the **natural** (or canonical) parameter.

Four ingredients, four jobs. The measure ν fixes what kind of thing the response is: Lebesgue measure on $\mathbb{R}$ for the normal, on $(0, \infty)$ for the gamma, counting measure on $\{0, 1, \dots\}$ for the Poisson. The cumulant function b determines every moment, as this section shows. The dispersion ϕ is a nuisance scale, unknown for the normal and the gamma, fixed at 1 for the Poisson and the binomial. The prior weight w is known and records how much data each observation summarizes: 1 for a single observation, m when y is an average of m of them. Only ϕ/w appears, so w never needs estimating; and that one formalism covers $w = 1$ and $w = m$ at once is what the common- b requirement buys, since it forces c to absorb the whole dependence on (ϕ, w) . That is what makes the family a *dispersion* family (Jørgensen 1987).

The function c does no statistical work: it absorbs whatever is left once the term linear in $y\theta$ has been extracted, and never depends on θ . That is the point, because in the log-likelihood

$$\log f(y; \theta, \phi) = \frac{w\{y\theta - b(\theta)\}}{\phi} + c(y, \phi, w), \quad (34.1.2)$$

the data and the parameter of interest meet only in the product $y\theta$. By Lemma 7.4.1, y itself is sufficient for θ .

The normalization $b = (\phi/w) \log \kappa$ is forced, because a density must integrate to one. The next lemma records the two properties of b that everything else rests on.

Lemma 34.1.1. The natural parameter space and the cumulant function

Θ is a convex subset of $\mathbb{R}$, and b is convex on Θ . If Y is not degenerate under some $\theta_0 \in \Theta$, then b is strictly convex on Θ .

Proof. Write $h(y) = e^{c(y, \phi, w)}$ and $s = w/\phi > 0$, so that $\kappa(\theta) = \int e^{sy\theta} h(y) d\nu(y)$. Let $\theta_1, \theta_2 \in \Theta$ and $\alpha \in (0, 1)$. Hölder's inequality with exponents $1/\alpha$ and $1/(1 - \alpha)$ applied to the factors $(e^{sy\theta_1} h)^\alpha$ and $(e^{sy\theta_2} h)^{1-\alpha}$ gives

$$\kappa(\alpha\theta_1 + (1 - \alpha)\theta_2) \leq \kappa(\theta_1)^\alpha \kappa(\theta_2)^{1-\alpha} < \infty.$$

So $\alpha\theta_1 + (1 - \alpha)\theta_2 \in \Theta$, which makes Θ convex, and taking logarithms shows $\log \kappa$, hence b , is convex. Equality in Hölder's inequality requires $e^{sy\theta_1} h$ and $e^{sy\theta_2} h$ to be proportional ν -almost everywhere, that is, $e^{sy(\theta_1 - \theta_2)}$ constant on the set where $h > 0$. For $\theta_1 \neq \theta_2$ this forces that set to be a single point, so Y is degenerate. ■

Throughout the chapter Θ is assumed to have nonempty interior and θ to lie in it — the same regularity Lemma 7.4.2 needs, and what makes the differentiations below legal.

34.1.2 Moments from the cumulant function**Proposition 34.1.2. Mean, variance and cumulants**

Let Y have density Eq. 34.1.1 with θ in the interior of Θ . Then Y has moments of all orders, and

- (a) $\mu := E(Y) = b'(\theta)$ and $\text{Var}(Y) = \frac{\phi}{w} b''(\theta)$;
- (b) the map $\theta \mapsto b'(\theta)$ is strictly increasing, hence a bijection from the interior of Θ onto an open interval $\mathcal{M}$, the **mean space**;
- (c) the cumulant generating function of Y is $K(t) = \log E e^{tY} = \frac{w}{\phi} \{b(\theta + t\phi/w) - b(\theta)\}$ for small $|t|$, so the r th cumulant of Y is $(\phi/w)^{r-1} b^{(r)}(\theta)$.

Proof. With h and $s = w/\phi$ as in the proof of Lemma 34.1.1, the density is $\kappa(\theta)^{-1} e^{sy\theta} h(y)$. The function κ is a two-sided Laplace transform of the measure $h d\nu$ evaluated at $-s\theta$; such a transform is analytic in the interior of its domain of convergence, and may be differentiated under the integral sign there, together with the finiteness of all moments (Widder 1941, chapter II; Brown 1986, chapter 2). Hence

$$\kappa'(\theta) = s \int y e^{sy\theta} h d\nu = s E(Y) \kappa(\theta), \quad \kappa''(\theta) = s^2 E(Y^2) \kappa(\theta).$$

Since $b = s^{-1} \log \kappa$, we have $b' = \kappa'/(s\kappa) = E(Y)$, which is (a) for the mean. Differentiating $\kappa' = s b' \kappa$ once more gives $\kappa'' = s b'' \kappa + s^2 (b')^2 \kappa$, so $E(Y^2) = b''/s + (b')^2$ and $\text{Var}(Y) = b''(\theta)/s = \phi b''(\theta)/w$.

- (b) By Lemma 34.1.1, b is strictly convex, so $b' = E(Y)$ is strictly increasing; it is contin-

uous, so its range on an open interval is an open interval.

- (c) For t small enough that $\theta + t/s \in \Theta$, $E e^{tY} = \kappa(\theta)^{-1} \int e^{(s\theta+t)y} h d\nu = \kappa(\theta + t/s)/\kappa(\theta)$; taking logarithms gives $K(t) = s^{-1}\{b(\theta + t/s) - b(\theta)\}$ with $s^{-1} = \phi/w$, and the coefficient of $t^r/r!$ in its expansion is $s^{-(r-1)}b^{(r)}(\theta)$. ■

Part (a) is the engine of the chapter: the variance is not free but determined by the mean, up to the single scalar ϕ/w . Part (b) lets us reparameterize by the mean, which is the parameter a modeller wants.

Definition 34.1.2. Variance function

For $\mu \in \mathcal{M}$ let $\theta(\mu)$ be the inverse of b' from Proposition 34.1.2(b). The **variance function** of the family is $V(\mu) = b''(\theta(\mu))$, so that

$$\text{Var}(Y) = \frac{\phi}{w} V(\mu), \quad \frac{d\theta}{d\mu} = \frac{1}{V(\mu)}.$$

The last identity is the derivative of an inverse function, $d\mu/d\theta = b'' = V$, and is used constantly below. The pair $(\mathcal{M}, V)$ determines the family: integrating V gives $\theta(\mu)$, hence b , hence the density.

Idea. A generalized linear model is specified by three separate choices: the **variance function** V , which fixes the family; the **link**; and the **linear predictor** $\mathbf{X}\beta$. Chapter 38 keeps the first two and throws the density away, which is possible because the estimating equations of Section 34.3 involve nothing else.

34.1.3 The standard families

The five classical families, plus the negative binomial with a known shape k , come from choosing ν and reading off b . The gamma row is written for shape a , so that $\phi = 1/a$. The measure ν is Lebesgue on $\mathbb{R}$ for the normal and on $(0, \infty)$ for the gamma and inverse Gaussian, counting measure on $\{0, 1, 2, \dots\}$ for the Poisson and negative binomial, and counting measure on $\{0, 1/m, \dots, 1\}$ for the binomial, whose response is a proportion out of m .

Family	$\theta(\mu)$	$b(\theta)$	ϕ	$V(\mu)$	$\mathcal{M}$
normal	μ	$\theta^2/2$	σ^2	1	$\mathbb{R}$
binomial	$\log \frac{\mu}{1-\mu}$	$\log(1 + e^\theta)$	1	$\mu(1 - \mu)$	$(0, 1)$
Poisson	$\log \mu$	e^θ	1	μ	$(0, \infty)$
gamma	$-1/\mu$	$-\log(-\theta)$	$1/a$	μ^2	$(0, \infty)$
inverse Gaussian	$-1/(2\mu^2)$	$-\sqrt{-2\theta}$	σ^2	μ^3	$(0, \infty)$
neg. binomial	$\log \frac{\mu}{\mu+k}$	$-k \log(1 - e^\theta)$	1	$\mu(1 + \mu/k)$	$(0, \infty)$

Two entries deserve a full derivation, because they show the two ways the prior weight and the

dispersion enter.

Example. The binomial as a family of proportions

Let $S \sim \text{Binomial}(m, \mu)$ and let $Y = S/m$ be the observed proportion. Its probability mass function at $y = s/m$ is

$$\begin{aligned} \binom{m}{my} \mu^{my} (1 - \mu)^{m(1-y)} \\ = \exp \left\{ m \left(y \log \frac{\mu}{1 - \mu} + \log(1 - \mu) \right) + \log \binom{m}{my} \right\}. \end{aligned}$$

This is [Eq. 34.1.1](#) with $w = m$, $\phi = 1$, $\theta = \log\{\mu/(1-\mu)\}$ and $b(\theta) = -\log(1-\mu) = \log(1+e^\theta)$. Then $b'(\theta) = e^\theta/(1+e^\theta) = \mu$ and $b''(\theta) = \mu(1-\mu)$, so [Proposition 34.1.2](#) gives $E(Y) = \mu$ and $\text{Var}(Y) = \mu(1-\mu)/m$: the number of trials acts entirely through the prior weight. The dispersion is fixed at 1, so a proportion more variable than $\mu(1-\mu)/m$ needs the different treatment of [Chapter 38](#).

Example. The gamma and its constant coefficient of variation

Let Y have the gamma density with shape $a > 0$ and mean μ ,

$$f(y) = \frac{(a/\mu)^a}{\Gamma(a)} y^{a-1} e^{-ay/\mu} \quad \text{on } (0, \infty).$$

Its logarithm is

$$a \left\{ y \left(-\frac{1}{\mu} \right) - \log \mu \right\} + (a-1) \log y + a \log a - \log \Gamma(a),$$

which is [Eq. 34.1.1](#) with $w = 1$, $\phi = 1/a$, $\theta = -1/\mu$ and $b(\theta) = \log \mu = -\log(-\theta)$. Then $b'(\theta) = -1/\theta = \mu$ and $b''(\theta) = 1/\theta^2 = \mu^2$, so $\text{Var}(Y) = \phi\mu^2$ and the coefficient of variation $\sqrt{\text{Var}(Y)}/\mu = \sqrt{\phi}$ is free of the mean: the spread is *proportional* to the level, the defining feature of the gamma family as a regression model. With $\mu = 4$ and $\phi = 0.4$ the formula gives $\text{Var}(Y) = 6.400$, and by [Proposition 34.1.2\(c\)](#) the third cumulant is $\phi^2 b'''(\theta) = 2\phi^2 \mu^3 = 20.480$; both agree with the exact gamma moments in the listing below.

```
import numpy as np
from scipy import stats

# Each family: theta(mu), b(theta) and its first three derivatives, and the exact
# distribution of Y, so that the exponential-family formulas can be checked.
FAMILIES = {
    "normal": dict(
        mu=2.0, phi=1.5, w=1.0,
        theta=lambda mu: mu,
        b=lambda t: t ** 2 / 2, b1=lambda t: t, b2=lambda t: 1.0 + 0 * t, b3=lambda
        t: 0 * t,
        dist=lambda mu, phi, w: stats.norm(mu, np.sqrt(phi / w)),
    ),
    "binomial": dict(                                     # Y = S/m, the sample
        proportion                                       proportion
        mu=0.3, phi=1.0, w=10.0,
```

```

        theta=lambda mu: np.log(mu / (1 - mu)),
        b=lambda t: np.log1p(np.exp(t)),
        b1=lambda t: 1 / (1 + np.exp(-t)),
        b2=lambda t: np.exp(-t) / (1 + np.exp(-t)) ** 2,
        b3=lambda t: np.exp(-t) * (np.exp(-t) - 1) / (1 + np.exp(-t)) ** 3,
        dist=None, # handled separately: Y is m
        ^-1 Binomial(m, mu)
    ),
    "Poisson": dict(
        mu=3.5, phi=1.0, w=1.0,
        theta=lambda mu: np.log(mu),
        b=lambda t: np.exp(t), b1=np.exp, b2=np.exp, b3=np.exp,
        dist=lambda mu, phi, w: stats.poisson(mu),
    ),
    "gamma": dict( # shape a = 1/phi
        mu=4.0, phi=0.4, w=1.0,
        theta=lambda mu: -1 / mu,
        b=lambda t: -np.log(-t), b1=lambda t: -1 / t, b2=lambda t: 1 / t ** 2,
        b3=lambda t: -2 / t ** 3,
        dist=lambda mu, phi, w: stats.gamma(a=w / phi, scale=mu * phi / w),
    ),
    "inverse Gaussian": dict(
        mu=2.0, phi=0.4, w=1.0,
        theta=lambda mu: -1 / (2 * mu ** 2),
        b=lambda t: -np.sqrt(-2 * t), b1=lambda t: (-2 * t) ** -0.5,
        b2=lambda t: (-2 * t) ** -1.5, b3=lambda t: 3 * (-2 * t) ** -2.5,
        dist=lambda mu, phi, w: stats.invgauss(mu=mu * phi / w, scale=w / phi),
    ),
}

def check(name, spec, k=None):
    """Compare b'(theta), phi b''(theta)/w and (phi/w)^2 b'''(theta) with the true
    moments."""
    mu, phi, w = spec["mu"], spec["phi"], spec["w"]
    t = spec["theta"](mu)
    mean_ef = spec["b1"](t)
    var_ef = phi * spec["b2"](t) / w
    cum3_ef = (phi / w) ** 2 * spec["b3"](t)
    if name == "binomial": # Y = S/m with S ~ Binomial(m
        , mu)
        m = int(w)
        s = np.arange(m + 1)
        p = stats.binom(m, mu).pmf(s)
        y = s / m
        mean, var = y @ p, (y - mu) ** 2 @ p
        cum3 = (y - mu) ** 3 @ p
    else:
        d = spec["dist"](mu, phi, w)
        mean, var = d.mean(), d.var()
        cum3 = d.stats("s") * var ** 1.5 # third cumulant = skewness *
        sd^3
    return mean_ef, mean, var_ef, var, cum3_ef, float(cum3)

for name, spec in FAMILIES.items():
    m_ef, m, v_ef, v, c_ef, c = check(name, spec)
    print(f"{name:17s} mean {m_ef:8.4f} {m:8.4f} var {v_ef:8.4f} {v:8.4f} cum3 {
c_ef:8.4f} {c:8.4f}")

```

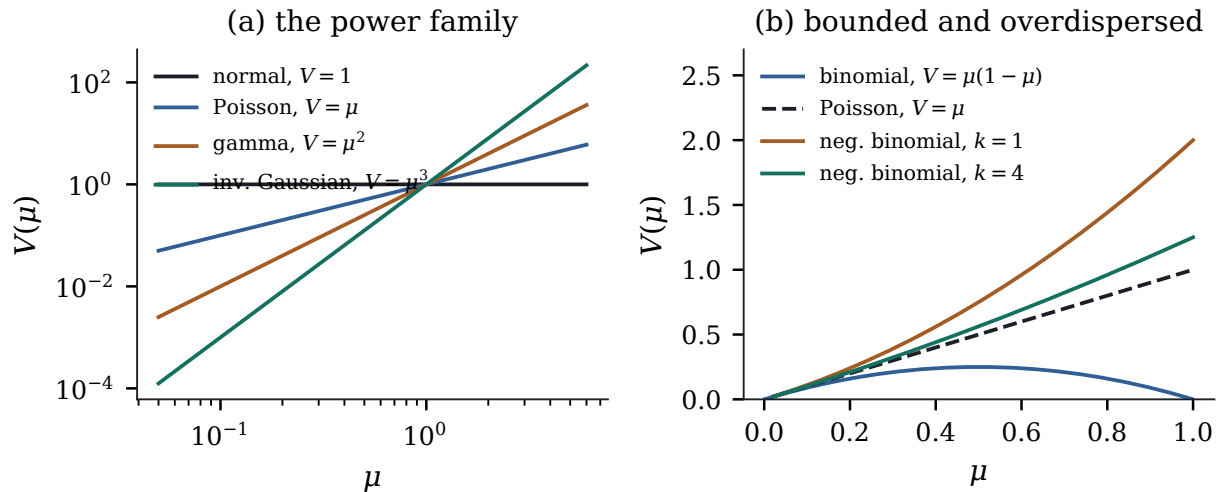


Figure 34.1.1: Variance functions. (a) The power family $V(\mu) = \mu^\xi$ on logarithmic axes: normal, Poisson, gamma and inverse Gaussian are $\xi = 0, 1, 2, 3$. (b) Two variance functions that are not powers. The binomial variance vanishes at both ends; the negative binomial variance $\mu(1 + \mu/k)$ lies above the Poisson line for every finite k .

[Run this code online](#)

The variance functions are drawn in [Figure 34.1.1](#). The normal, Poisson, gamma and inverse Gaussian are powers, $V(\mu) = \mu^\xi$ with $\xi = 0, 1, 2, 3$; interpolating between them gives a continuum of families (Jørgensen 1987) that includes a useful model for a response with a point mass at zero and a continuous positive part. The remaining two are not powers: the binomial variance vanishes at both ends of $(0, 1)$, because a proportion pinned near 0 or 1 cannot vary, and the negative binomial variance exceeds the Poisson variance by an amount set by k .

Example. The negative binomial with known shape

Let $\Pr(Y = y) = \frac{\Gamma(y+k)}{\Gamma(k)y!} \pi^y (1-\pi)^k$, $y = 0, 1, 2, \dots$, for a fixed $k > 0$. Writing $\theta = \log \pi$ puts this in the form [Eq. 34.1.1](#) with $w = 1$, $\phi = 1$ and $b(\theta) = -k \log(1 - e^\theta)$, so that

$$\mu = b'(\theta) = \frac{k\pi}{1-\pi}, \quad b''(\theta) = \frac{k\pi}{(1-\pi)^2} = \mu \left(1 + \frac{\mu}{k}\right)$$

after substituting $\pi = \mu/(\mu + k)$. With $k = 2$ and $\mu = 3$ this gives $\text{Var}(Y) = 7.500$, against 3 for a Poisson variable with the same mean.

The caveat is essential: this is an exponential dispersion family **only for fixed** k . The shape enters the variance function, not the dispersion, so it cannot be divided out, and letting k vary leaves the form [Eq. 34.1.1](#). Estimating k is outside this chapter's machinery; [Chapter 37](#) takes it up.

34.1.4 Sufficiency, and what the natural parameter buys

Suppose $Y_1, \dots, Y_n$ are independent, each with density [Eq. 34.1.1](#), with a common b and ϕ , known weights w_i , and natural parameters θ_i . The joint log-likelihood is

$$\ell = \frac{1}{\phi} \sum_{i=1}^n w_i \{y_i \theta_i - b(\theta_i)\} + \sum_{i=1}^n c(y_i, \phi, w_i). \quad (34.1.3)$$

Nothing has been assumed yet about how the θ_i depend on regressors — that is the next section's business — but [Eq. 34.1.3](#) already shows what to expect. If the θ_i are constrained to a p -dimensional linear family $\theta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$, then $\sum_i w_i y_i \theta_i = \boldsymbol{\beta}^\top \mathbf{X}^\top \text{diag}(\mathbf{w}) \mathbf{y}$ and the data enter the likelihood only through the p numbers $\mathbf{X}^\top \text{diag}(\mathbf{w}) \mathbf{y}$. With ϕ known the remaining term $\sum_i c(y_i, \phi, w_i)$ is a function of the data alone, so Lemma 7.4.1 makes this vector sufficient for $\boldsymbol{\beta}$, and if the attainable $\boldsymbol{\beta}$ contain an open subset of $\mathbb{R}^p$, Lemma 7.4.2 makes it complete. With ϕ unknown that term depends on a parameter, both arguments fail, and a second statistic is needed — as in the normal linear model, the case $b(\theta) = \theta^2/2$, $w_i = 1$, where $\mathbf{X}^\top \mathbf{y}$ is accompanied by $\mathbf{y}^\top \mathbf{y}$ for σ^2 (Proposition 7.4.3).

That is a genuine reduction, and it is why the natural parameter deserves its name. What it does not give is any analogue of the Gauss–Markov theorem: the maximum likelihood estimate is a nonlinear function of $\mathbf{y}$, generally biased, with an exact distribution unknown outside the normal case. Everything finite-sample that Parts II and III delivered is gone, and [Section 34.5](#) replaces it with asymptotics.

Warning. The reduction needs θ_i — not μ_i , and not some other function of it — to be linear in $\boldsymbol{\beta}$. That is the *canonical link*, one modelling choice among several. With any other link the reduction fails and the data enter the likelihood through the whole vector $\mathbf{y}$; [Section 34.2](#) explains why one might accept that.

34.1.5 Exercises

34.1.5.1 A. Check your understanding

Exercise. A1

Write the Poisson probability mass function in the form [Eq. 34.1.1](#), identify θ , b , ϕ , w and c , and verify $E(Y) = \text{Var}(Y) = \mu$ using [Proposition 34.1.2](#). What is Θ ?

Solution. $\Pr(Y = y) = e^{-\mu} \mu^y / y! = \exp\{y \log \mu - \mu - \log y!\}$, so $\theta = \log \mu$, $b(\theta) = e^\theta$, $\phi = w = 1$ and $c(y) = -\log y!$. Then $b' = b'' = e^\theta = \mu$, and [Proposition 34.1.2](#) gives $E(Y) = \text{Var}(Y) = \mu$. The sum $\sum_y e^{y\theta} / y! = e^{e^\theta}$ converges for every real θ , so $\Theta = \mathbb{R}$ and $\mathcal{M} = (0, \infty)$.

34.1.5.2 B. Practice

Exercise. B1

Verify the inverse Gaussian row of the table: starting from

$$f(y) = (2\pi\sigma^2 y^3)^{-1/2} \exp\left\{-\frac{(y - \mu)^2}{2\sigma^2 \mu^2 y}\right\} \quad \text{on } (0, \infty),$$

put it in the form [Eq. 34.1.1](#) and confirm $V(\mu) = \mu^3$. What is Θ , and what happens at $\theta = 0$?

Solution. Expanding the square, $-(y - \mu)^2/(2\sigma^2\mu^2y) = -y/(2\sigma^2\mu^2) + 1/(\sigma^2\mu) - 1/(2\sigma^2y)$. So the exponent is $\sigma^{-2}\{y\theta - b(\theta)\}$ with $\theta = -1/(2\mu^2)$ and $b(\theta) = -1/\mu = -\sqrt{-2\theta}$, while the remaining terms $-\frac{1}{2}\log(2\pi\sigma^2y^3) - 1/(2\sigma^2y)$ make up $c(y, \phi, w)$ with $\phi = \sigma^2$, $w = 1$. Then $b'(\theta) = (-2\theta)^{-1/2} = \mu$ and $b''(\theta) = (-2\theta)^{-3/2} = \mu^3$. Here $\Theta = (-\infty, 0]$ and $\theta = 0$ is a boundary point: the integral still converges there, but b' blows up, so $\mathcal{M} = (0, \infty)$ is the image of the *interior* only.

Exercise. B2

Show that the variance function determines the family: if V is given on an interval $\mathcal{M}$, then $\theta(\mu) = \int^\mu dt/V(t)$ and $b(\theta(\mu)) = \int^\mu t dt/V(t)$, each up to constants that only relocate the density. Carry this out for $V(\mu) = \mu(1 - \mu)$ on $(0, 1)$ and recover the binomial row.

Solution. By [Definition 34.1.2](#), $d\theta/d\mu = 1/V(\mu)$, which integrates to the stated $\theta(\mu)$. Also $db/d\mu = (db/d\theta)(d\theta/d\mu) = \mu/V(\mu)$, giving the second integral. For $V(\mu) = \mu(1 - \mu)$, partial fractions give $\int d\mu/\{\mu(1 - \mu)\} = \log\{\mu/(1 - \mu)\} = \theta$, and $\int \mu d\mu/\{\mu(1 - \mu)\} = -\log(1 - \mu) = \log(1 + e^\theta) = b(\theta)$, which is the binomial row.

Exercise. B3

Use [Proposition 34.1.2\(c\)](#) to show that the skewness of Y is $\gamma_1 = \sqrt{\phi/w} V'(\mu)/\sqrt{V(\mu)}$. Evaluate it for the Poisson, the gamma and the binomial, and say in each case how the skewness behaves as the mean moves.

Solution. The third cumulant is $(\phi/w)^2 b'''(\theta)$, and $b'''(\theta) = dV/d\theta = V'(\mu) d\mu/d\theta = V'(\mu)V(\mu)$. Dividing by $\text{Var}(Y)^{3/2} = \{(\phi/w)V\}^{3/2}$ gives $\gamma_1 = \sqrt{\phi/w} V'(\mu)/\sqrt{V(\mu)}$. For the Poisson ($\phi = w = 1$, $V = \mu$), $\gamma_1 = \mu^{-1/2}$: small counts are strongly skewed and large ones nearly symmetric. For the gamma, $V = \mu^2$ gives $\gamma_1 = 2\sqrt{\phi}$, free of the mean. For the binomial with $w = m$, $\gamma_1 = (1 - 2\mu)/\sqrt{m\mu(1 - \mu)}$, which changes sign at $\mu = \frac{1}{2}$.

34.1.5.3 C. Going deeper

Exercise. C1

Show that the uniform family on $(0, \vartheta)$ is not an exponential dispersion family for any choice of ν , and that the Cauchy family with median ϑ is not either. Which hypothesis of [Definition 34.1.1](#) fails in each case?

Exercise. C2

Suppose $V(\mu) = \mu^\xi$ on $(0, \infty)$ for some $\xi \notin \{1, 2\}$. Carry out [Exercise \(B2\)](#) to find $\theta(\mu)$

and $b(\theta)$, and check that $\xi = 0$ and $\xi = 3$ reproduce the normal and inverse Gaussian rows. For $1 < \xi < 2$ the family is the distribution of a Poisson number of independent gamma variables, with positive probability at 0 and a continuous part on $(0, \infty)$ (Jørgensen 1987); confirm from the general formula that its mean space is $(0, \infty)$.